

CIPHER SHIFT

Dynamic Length-Based Encryption Prototype

Industrial Cryptography Presentation // v1.0

THE ROTATION PARADIGM

LOGIC ARCHITECTURE

Cipher Shift is a customized Caesar Cipher implementation where the shift value is **not static**. Instead, it is derived from the first segment of the injected payload.

- Variable word-length dependency.
- Synchronized matrix mapping.
- Real-time alphanumeric rotation.



TECHNICAL INFRASTRUCTURE



FLASK BACKEND

Micro-service architecture handling payload processing and rotation calculations via Python.



NATIVE ES6+

Asynchronous frontend logic for immersive UI hooks, particle effects, and DOM manipulation.



TAILWIND CSS

Utility-first styling framework used for the glassmorphism panels and industrial layouts.

ALGORITHMIC SHIFT

MOD

26

Circular Alpha Logic

PAYLOAD NORMALIZATION

The backend identifies the first continuous string segment.
Its length L defines the shift factor S .

$$C = (P + S) \bmod 26$$



IMMERSIVE UX

"BREAKTHESEAL"

To move beyond generic web styles, the UI implements **mechanical hooks**. Outputs are locked behind a manual override firewall, forcing human interaction before decryption occurs.

- [+] CRT Scanline Filters
- [+] Kinetic Glyph Strips
- [+] High-Voltage Zap Effects

PROJECT HIERARCHY

```
cipher_project/  
├── app.py (Main Logic)  
├── static/  
│   ├── css/ (Input.css)  
│   └── js/ (Interactivity)  
└── templates/  
    └── index.html (UI)
```

MODULE ROLES

File	Description
app.py	Flask Server & Route logic.
index.html	The core Brutalist interface.
static/js	Asynchronous matrix execution.

QUESTIONS?

System Standby // Awaiting Human Input

```
ROOT@CIPHER_SHIFT:~$ _
```


IMAGE SOURCES



https://images.stockcake.com/public/f/6/5/f6555150-07c3-4a98-b0a2-e8e147c5f07a_large/neon-pixel-interface-stockcake.jpg

Source: stockcake.com



https://images.stockcake.com/public/5/f/4/5f4836e7-0397-44e5-8c63-ac8113b48921_large/digital-dna-discovery-stockcake.jpg

Source: stockcake.com